Simultaneous Linear Algebraic Equations

Jean Paul Nsabimana,
Department of Mathematics
University of Rwanda
Email: nsapa123@gmail.com

April 13, 2026

Outline

1. Simultaneous Linear Algebraic Equations
   - Direct Method
     - Gauss Elimination Method
     - Gauss - Jordan Method
   - Iterative Method
     - Jacobi Method of iteration
     - Gauss-Sidel Iteration Method

Introduction

- Simultaneous linear algebraic equations are very common in various fields of engineering and science. We use matrix inversion method or cramer's rule to solve these equations in general.
- To solve the equations contain a large number of unknowns; Particularly suited for computer operations.
- These are of two types:
  1. Direct Method;
  2. Iterative methods.

Gauss Elimination Method

- In this method, the unknowns are eliminated successively by transforming the given matrix into an equivalent system with upper triangular coefficient matrix (i.e a matrix in which all the elements below the principal diagonal are zero) by means of elementary row operations, from which the unknowns are found by back substitution.
- Here, we shall explain it by considering of three equations in three unknowns.

Gauss Elimination Method

Consider a system

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases} \tag{1}$$

where x, y, z are unknowns. The system in matrix form is

$$AX = B,$$

where

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_{13} \end{bmatrix}, \qquad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \qquad \textbf{and} \qquad B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}.$$

Gauss Elimination Method

Consider the augmented matrix $[A|b]$,

$$[A|B] \quad = \quad \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_{13} & d_3 \end{bmatrix} \tag{2}$$

Now, Equation (2) is to be reduced to an upper triangular matrix, let $a_1 \neq 0$. Then

$$\begin{cases} R_2 \Rightarrow R_2 - \frac{a_2}{a_1} R_1 \\ R_3 \Rightarrow R_3 - \frac{a_3}{a1} R_1 \end{cases} \quad \Leftrightarrow \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ 0 & b_2' & c_2' & d_2' \\ 0 & b_3' & c_{13}' & d_3' \end{bmatrix} \tag{3}$$

Here, a_1 is called the first pivot and $b_2', c_2', b_3', c_3', d_2', d_2'$ are transformed elements. Now take b_2' as the pivot $(b'2 \neq o)$, then

Gauss Elimination Method

$$R_3 \Rightarrow R_3 - \frac{b_3'}{b_2'} R_2 \Leftrightarrow \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ 0 & b_2' & c_2' & d_2' \\ 0 & 0 & c_{13}'' & d_3'' \end{bmatrix} \tag{4}$$

Now, if $c_3'' \neq 0$, from Equation (4). The given system of linear equation is equivalent to

$$\begin{cases} a_1 x + b_1 y + c_1 z = d_1 \\ b_2^{'} y + c_2^{'} z = d_2^{'} \\ c_3^{''} z = d_3^{''}. \end{cases} \tag{5}$$

Using back substitution,

$$\begin{cases} z = \frac{d_3^{''}}{c_3^{''}} \\ y = \frac{1}{b_2^{'} c_3^{''}} \left(d_2^{'} c_3^{''} - c_2^{'} d_3^{''} \right) \\ x = \frac{1}{a_1 b_2^{'} c_3^{''}} \left(d_1 b_2^{'} c_3^{''} - b_1 d_2^{''} c_3^{''} + b_1 c_2^{'} d_3^{''} - b_2^{'} c_1 d_3^{''} \right) \end{cases} \tag{6}$$

Note and example

Note: This method fails if any one of the pivots $a_1, b_2^{'}$ or $c_3^{''}$ becomes zero. In such case, by interchanging the rows we can get the non-zero pivots.

Example

Solve the system of equations

$$\begin{cases} 3x + y - z = 3 \\ 2x - 8y + z = -5 \\ x - 2y + 9z = 8 \end{cases},$$

using Gauss elimination method.

Solution

The given system is equivalent to

$$\begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \\ 8 \end{bmatrix}$$

The augmented matrix is

$$[A|B] = \begin{bmatrix} 3 & 1 & -1 & 3 \\ 2 & -8 & 1 & -5 \\ 1 & -2 & 9 & 8 \end{bmatrix}$$

Now we will make A as upper triangular choosing 3 as a pivot,

$$\begin{cases} R_2 \Rightarrow R_2 - \frac{2}{3}R_1 \\ R_3 \Rightarrow R_3 - \frac{1}{3}R_1 \end{cases} \Leftrightarrow \begin{bmatrix} 3 & 1 & -1 & 3 \\ 0 & -\frac{26}{3} & \frac{5}{3} & -7 \\ 0 & -\frac{7}{3} & \frac{28}{3} & 7 \end{bmatrix}$$

Solution Cont...

Now choosing $-\frac{26}{3}$ as the pivot from the second column, we get

$$R_3 \Rightarrow R_3 - \frac{7}{26}R_2 \qquad \Leftrightarrow \begin{bmatrix} 3 & 1 & -1 & 3 \\ 0 & -\frac{26}{3} & \frac{5}{3} & -7 \\ 0 & 0 & \frac{693}{78} & \frac{231}{26} \end{bmatrix}$$

From this, we get

$$\begin{cases} 3x + y - z = 3 \\ -\frac{26}{3}y + \frac{5}{3}z = -7 \\ \frac{693}{78}z = \frac{231}{26} \end{cases} \qquad \Rightarrow \begin{cases} z = 1 \\ -\frac{26}{3}y = -7 + \frac{5}{3} = -\frac{26}{3} \Rightarrow y = 1 \\ x = \frac{1}{3}(3 - y + z) = \frac{1}{3}(3 - 1 + 1) = 1 \end{cases}$$

$$\therefore \begin{cases} z = 1 \\ y = 1 \\ x = 1 \end{cases}$$

Exercise IV.1

1. Solve the system of equations

$$\begin{cases} 28x + 4y - z = 32 \\ x + y + 10z = 24 \\ 2x + 17y + 4z = 35 \end{cases}$$

By Gauss elimination method.

2. Using Gauss elimination method. Solve the system of equations

$$\begin{cases} 3.15x - 1.96y + 3.85z = 12.95 \\ 2.13x + 5.12y - 2.89z = -8.61 \\ 5.92x + 3.05y + 2.15z = 6.88 \end{cases}$$

3. Solve the system of equations

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 2 \\ x_1 + x_2 + 3x_3 - 2x_4 = -6 \\ 2x_1 + 3x_2 - x_3 + 2x_4 = 7 \\ x_1 + 2x_2 + x_3 - x_4 = -2 \end{cases}$$

Gauss - Jordan Method

- This method is a modified form of Gauss elimination method. In this method, the coefficient matrix A of $AX = B$ is reduced to a diagonal matrix or unit matrix by making all the elements above and below to the principal diagonal of A zero.
- The labor of back substitution is saved here even though it involves additional computation.

Example

Solve the system of equations

$$\begin{cases} 10x + y + z = 12 \\ 2x + 10y + z = 13 \\ x + y + 5x = 7 \end{cases} \qquad \text{by Gauss - Jordan method.}$$

Solution

The given system in matrix form is

$$\begin{bmatrix}10 & 1 & 1\\ 2 & 10 & 1\\ 1 & 1 & 5\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}12\\ 13\\ 7\end{bmatrix} \qquad \therefore [A|B]=\begin{bmatrix}10 & 1 & 1 & 12\\ 2 & 10 & 1 & 13\\ 1 & 1 & 5 & 7\end{bmatrix}.$$

$$R_1 \to R_1 - 9R_3 \quad \Leftrightarrow \quad \begin{bmatrix}1 & -8 & -44 & -51\\ 2 & 10 & 1 & 13\\ 1 & 1 & 5 & 7\end{bmatrix}$$

$$\begin{cases}R_2 \to R_2 - 2R_1\\ R_3 \to R_3 - R_1\end{cases} \quad \Leftrightarrow \quad \begin{bmatrix}1 & -8 & -44 & -51\\ 0 & 26 & 89 & 115\\ 0 & 9 & 49 & 58\end{bmatrix}$$

$$R_2 \to -(R_2 - 3R_3) \quad \Leftrightarrow \quad \begin{bmatrix}1 & -8 & -44 & -51\\ 0 & 1 & 58 & 59\\ 0 & 9 & 49 & 58\end{bmatrix}$$

Solution

$$\begin{cases} R_1 \to R_1 + 8R_2 \\ R_3 \to R_3 - 9R_2 \end{cases} \Leftrightarrow \begin{bmatrix} 1 & 0 & 420 & 421 \\ 0 & 1 & 58 & 59 \\ 0 & 0 & -473 & -473 \end{bmatrix}$$

$$R_3 \to -\frac{1}{473}R_3 \Leftrightarrow \begin{bmatrix} 1 & 0 & 420 & 421 \\ 0 & 1 & 58 & 59 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{cases} R_1 \to R_1 - 420R_3 \\ R_2 \to R_2 - 58R_3 \end{cases} \Leftrightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Therefore, the system $AX = B$ reduced to the form

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad \therefore \begin{cases} x = 1 \\ y = 1 \\ z = 1 \end{cases}$$

Exercise IV.2

1. Solve the system of equations

$$\begin{cases} 10x_1 + x_2 + x_3 = 12, \\ x_1 + 10x_2 - x_3 = 10, \\ x_1 - 2x_2 + 10x_3 = 9, \end{cases} \qquad \text{by Gauss - Jordan method.}$$

2. Solve the following system of equations by Gauss - Jordan method

$$\begin{cases} x + 2y + z - w = -2, \\ 2x + 3y - z + 2w = 7, \\ x + y + 3z - 2w = -6, \\ x + y + z + w = 2. \end{cases}$$

Introduction

- The method of iteration is not applicable to all systems of equations. For this, **each equation of the system must contain one large coefficient and large coefficient must be attached to a different unknown in that equation**.
- The solution to a system of linear equations will exist by iterative procedure if the absolute value of largest coefficient is greater than the sum of the absolute values of all remaining coefficients in each equation. (i.e. **This is the condition for convergence**).
- Now let us see some of such method in details:

Jacobi Method of iteration

- This method is also known as **Gauss-Jacobi method.**

$$
\begin{aligned}
a_1x + b_1y + c_1z &= d_1 \\
a_2x + b_2y + c_2z &= d_2 \\
a_3x + b_3y + c_3z &= d_3
\end{aligned}
\tag{7}
$$

- Let $|a_1| > |b_1| + |c_1|; \quad |b_2| > |a_2| + |c_2|; \quad |c_3| > |a_3| + |b_3|.$
- That is, in each equation the coefficients of diagonal terms are large. Hence the system (8) is ready for iteration. Solving for x, y and z respectively, we get

$$
\begin{aligned}
x &= \frac{1}{a_1}(d_1 - b_1y - c_1z) \\
y &= \frac{1}{b_2}(d_2 - a_2x - c_2z) \\
z &= \frac{1}{c_3}(d_3 - a_3x - b_3y)
\end{aligned}
\tag{8}
$$

Jacobi Method of iteration

Let x_0, y_0 and z_0 be the initial approximations of x, y and z. Substituting these values on RHS of Equation (9), the first approximations:

$$
\begin{aligned}
x_1 &= \frac{1}{a_1}(d_1 - b_1 y_0 - c_1 z_0) \\
y_1 &= \frac{1}{b_2}(d_2 - a_2 x_0 - c_2 z_0) \\
z_1 &= \frac{1}{c_3}(d_3 - a_3 x_0 - b_3 y_0)
\end{aligned} \tag{9}
$$

Substituting the values of x_1, y_1 and z_1 in the RHS of Equation (9), the second approximations are given by

$$
\begin{aligned}
x_2 &= \frac{1}{a_1}(d_1 - b_1 y_1 - c_1 z_1) \\
y_2 &= \frac{1}{b_2}(d_2 - a_2 x_1 - c_2 z_1) \\
z_2 &= \frac{1}{c_3}(d_3 - a_3 x_1 - b_3 y_1)
\end{aligned} \tag{10}
$$

Jacobi Method of iteration

Proceeding in the same way, if $x_r, \quad y_r, \quad z_r$ are the r^{th} iterates, then

$$\begin{aligned} x_{r+1} &= \frac{1}{a_1}(d_1 - b_1 y_r - c_1 z_r) \\ y_{r+1} &= \frac{1}{b_2}(d_2 - a_2 x_r - c_2 z_r) \\ z_{r+1} &= \frac{1}{c_3}(d_3 - a_3 x_r - b_3 y_r) \qquad (11) \end{aligned}$$

The process continued till convergence is secured.
Note: In the absence of any better estimates, the initial approximations are taken as $x_0 = y_0 = z_0 = 0$.

Example

Solve by Jacobi iteration method the system

$$\begin{aligned} 8x - 3y + 2z &= 20; \\ 6x + 3y + 12z &= 35; \\ 4x + 11y - z &= 33. \end{aligned}$$

Solution: Consider the given system as

$$\begin{aligned} 8x - 3y + 2z &= 20; \\ 4x + 11y - z &= 33; \\ 6x + 3y + 12z &= 35. \end{aligned}$$

Solution

So that the diagonal elements are **dominant in the coefficient matrix.** Now we write the equations in the form

$$\begin{aligned} x &= \frac{1}{8}(20+3y-2z) \\ y &= \frac{1}{11}(33-4x+z) \\ z &= \frac{1}{12}(35-6x-3y) \end{aligned} \tag{12}$$

We start from an approximations $x_0 = y_0 = z_0 = 0$. Substituting these values on RHS of equation (12). First approximation is given as:

$$\begin{aligned} x_1 &= \frac{1}{8}(20+3y_0-2z_0) = 2.5 \\ y_1 &= \frac{1}{11}(33-4x_0+z_0) = 3 \\ z_1 &= \frac{1}{12}(35-6x_0-3y_0) = 2.9166667 \end{aligned}$$

Solution

Second approximation: substituting x_1, y_1, z_1 on RHS of Equations (12), we get

$$\begin{aligned} x_2 &= \frac{1}{8}(20 + 3y_1 - 2z_1) = 2.895833 \\ y_2 &= \frac{1}{11}(33 - 4x_1 + z_1) = 2.3560606 \\ z_2 &= \frac{1}{12}(35 - 6x_1 - 3y_1) = 0.9166666 \end{aligned}$$

Third approximation: Substituting x_2, y_2, z_2 on RHS of Equation (12), we get

$$\begin{aligned} x_3 &= \frac{1}{8}(20 + 3y_2 - 2z_2) = 3.1543561 \\ y_3 &= \frac{1}{11}(33 - 4x_2 + z_2) = 2.030303 \\ z_3 &= \frac{1}{12}(35 - 6x_2 - 3y_2) = 0.8797348 \end{aligned}$$

Solution

Proceeding in the same way, we get the following results

$$\begin{aligned} x_4 &= 3.0419299 & x_5 &= 3.0168733 & x_6 &= 3.0104409 & x_7 &= 3.0157694 \\ y_4 &= 1.9329373 & y_5 &= 1.9696539 & y_6 &= 1.9859295 & y_7 &= 1.9885503 \\ z_4 &= 0.8319128 & z_5 &= 0.9127173 & z_6 &= 0.9158165 & z_7 &= 0.9149638 \end{aligned}$$

Until

$$\begin{aligned} x_{11} &= 3.\underline{0167}368 & x_{12} &= 3.\underline{0167}424 \\ y_{11} &= 1.\underline{9858}681 & y_{12} &= 1.\underline{9858}987 \\ z_{11} &= 0.\underline{9118}326 & x_{12} &= 0.\underline{9118}312 \end{aligned}$$

Therefore, from 11^{th} and 12^{th} approximations, the values of x, y, z are same correct to four decimal places. Stopping at this stage, we get

$$\begin{aligned} x &= 3.0167; \\ y &= 1.9858; \\ z &= 0.9118. \end{aligned}$$

Exercise IV.3

Solve the following system of equations by Jacobi iteration method:

$$\begin{aligned} 3x + 4y + 15z &= 54.8; \\ x + 12y + 3z &= 39.66; \\ 10x + y - 2z &= 7.74. \end{aligned}$$

Gauss-Sidel Iteration Method

This is a modification of Gauss - Jacobi method. As before, the system of linear equations

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \tag{13}$$

is written as

$$\begin{aligned} x &= \frac{1}{a_1}(d_1 - b_1y - c_1z) \quad (a) \\ y &= \frac{1}{b_2}(d_2 - a_2x - c_2z) \quad (b) \\ z &= \frac{1}{c_3}(d_3 - a_3x - b_3y) \quad (c) \end{aligned} \tag{14}$$

We start with the initial approximations x_0, y_0, z_0. Substituting y_0 and z_0 in equation (a),

Gauss-Sidel Iteration Method

We get

$$x_1 = \frac{1}{a_1}(d_1 - b_1 y_0 - c_1 z_0).$$

Now substituting $x = x_1, z = z_0$ in Equation (b), we get

$$y_1 = \frac{1}{b_2}(d_2 - a_2 x_1 - c_2 z_0).$$

Now substituting $x = x_1, y = y_1$ in Equation (c), we get

$$z_1 = \frac{1}{c_3}(d_3 - a_3 x_1 - b_3 y_1).$$

Gauss-Sidel Iteration Method

This process is continued till the values of x, y, z are obtained to the desired degree of accuracy. The general algorithm is as follows:

$$
\begin{aligned}
x_{k+1} &= \frac{1}{a_1}(d_1 - b_1 y_k - c_1 z_k) \\
y_{k+1} &= \frac{1}{b_2}(d_2 - a_2 x_{k+1} - c_2 z_k) \\
z_{k+1} &= \frac{1}{c_3}(d_3 - a_3 x_{k+1} - b_3 y_{k+1})
\end{aligned} \tag{15}
$$

Since the current values of the unknowns at each stage of iteration are used in proceeding to the next stage of iteration, this method is more rapid in convergence than Gauss- Jacobi Method.

Example

Solve by Gauss-Seidal iteration method the system

$$\begin{aligned} 8x - 3y + 2z &= 20; \\ 4x + 11y - z &= 33; \\ 6x + 3y + 12z &= 35. \end{aligned}$$

So that the diagonal elements are **dominant in the coefficient matrix.** Now we write the equations in the form

$$\begin{aligned} x &= \frac{1}{8}(20 + 3y - 2z) \quad (a) \\ y &= \frac{1}{11}(33 - 4x + z) \quad (b) \\ z &= \frac{1}{12}(35 - 6x - 3y) \quad (c) \end{aligned}$$

Solution

Putting $y = 0$, $z = 0$ in RHS of (a), we get $x_1 = \frac{20}{8} = 2.5$. Putting $x = 2.5$, $z = 0$ in RHS of (b), we get

$$y_1 = \frac{1}{11}(33 - 4(2.5)) = 2.0909091.$$

Putting $x = 2.5$, $y = 2.0909091$ in RHS of (c), we get

$$z_1 = \frac{1}{12}(35 - 6(2.5) - 3(2.0909091)) = 1.1439394.$$

For second approximation:

$$\begin{aligned}
x_2 &= \frac{1}{8}(20 + 3y_1 - 2z_1) = 2.9981061 \\
y_2 &= \frac{1}{11}(33 - 4x_2 + z_1) = 2.0137741 \\
z_2 &= \frac{1}{12}(35 - 6x_2 - 3y_2) = 0.9141701
\end{aligned}$$

Solution

For third approximation:

$$\begin{aligned} x_3 &= \frac{1}{8}(20 + 3y_2 - 2z_2) = 3.0266228 \\ y_3 &= \frac{1}{11}(33 - 4x_3 + z_2) = 1.9825163 \\ z_3 &= \frac{1}{12}(35 - 6x_3 - 3y_3) = 0.9077262 \end{aligned}$$

For fourth approximation:

$$\begin{aligned} x_4 &= \frac{1}{8}(20 + 3y_3 - 2z_3) = 3.0165121 \\ y_4 &= \frac{1}{11}(33 - 4x_4 + z_3) = 1.9856071 \\ z_4 &= \frac{1}{12}(35 - 6x_4 - 3y_4) = 0.9120088 \end{aligned}$$

Solution

For fifth approximation:

$$
\begin{aligned}
x_5 &= \frac{1}{8}(20 + 3y_4 - 2z_4) = 3.0166005 \\
y_5 &= \frac{1}{11}(33 - 4x_5 + z_4) = 1.9859643 \\
z_5 &= \frac{1}{12}(35 - 6x_5 - 3y_5) = 0.9118753
\end{aligned}
$$

For sixth approximation:

$$
\begin{aligned}
x_6 &= \frac{1}{8}(20 + 3y_5 - 2z_5) = 3.0167678 \\
y_6 &= \frac{1}{11}(33 - 4x_6 + z_5) = 1.9858913 \\
z_6 &= \frac{1}{12}(35 - 6x_6 - 3y_6) = 0.9118099
\end{aligned}
$$

Solution

For seventh approximation:

$$
\begin{aligned}
x_7 &= \frac{1}{8}(20 + 3y_6 - 2z_6) = 3.0167568 \\
y_7 &= \frac{1}{11}(33 - 4x_7 + z_6) = 1.9858894 \\
z_7 &= \frac{1}{12}(35 - 6x_7 - 3y_7) = 0.9118159
\end{aligned}
$$

Since at the sixth and seventh approximations, the values of x, y, z are the same, correct to four decimal places; We can stop the iteration process. Therefore

$$
\begin{aligned}
x &= 3.0167 \\
y &= 1.9858 \\
z &= 0.9118
\end{aligned}
$$

We find that 12 iterations are necessary in Gauss - Jacobi method to get the same accuracy as achieved by 7 iterations in Gauss - Seidel method.

Exercise IV.4

1. Solve by Gauss- Seidel method the following system of equations

$$\begin{aligned} 28x + 4y - z &= 32 \\ x + 3y + 10z &= 24 \\ 2x + 17y + 4z &= 35 \end{aligned}$$

2. Using Gauss-Seidel iteration method, find the solution of the following system of equations correct to four decimal places.

$$\begin{aligned} 10x - 2y - z - w &= 3 \\ -2x + 10y - z - w &= 15 \\ -x - y + 10z - 2w &= 27 \\ -x - y - 2z + 10w &= -9 \end{aligned}$$

Exercise IV.4

3. Solve the following system of linear equations by

1. Gauss-Jacobi method
2. Gauuss-seidel iteration method

(i)

$$\begin{aligned} 2x + y + z &= 4 \\ x + 2y + z &= 4 \\ x + y + 2z &= 4 \end{aligned}$$

(ii)

$$\begin{aligned} 8x + y + z &= 8 \\ 2x + 4y + z &= 4 \\ x + 3y + 5z &= 5 \end{aligned}$$